Causal sufficient dimension reduction for multiple continuous exposures

An application to environmental mixtures analysis

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Introduction

Existing approaches force a difficult tradeoff

- Single-index methods emphasize interpretability.
- Full exposure-response methods emphasize flexibility.
- **Causal SDR** targets a low-dimensional causal representation.

The target is a causal subspace

Definition

We seek a matrix $\beta \in \mathbb{R}^{p \times d}$ such that

$$\mu(x) = \mu_\beta(\beta^\top x), \qquad d \ll p.$$

Key implication

The intervention-response surface depends on x only through $\beta^\top x$.

Results

Causal estimators recover the target subspace

Method	Adjusts for confounding	Target
MAVE	No	Association subspace
RA-MAVE	Yes	Causal subspace
DR-MAVE	Yes	Causal subspace

The first-stage error appears explicitly

$$\left\|(I - \widehat{\beta}\widehat{\beta}^{\top})\beta_0\right\| = O_p\left(h^3 + h\delta_n + h^{-1}\delta_n^2 + h^{-1}R_n\right).$$

Gold is reserved for the term being discussed; red is reserved for genuine warnings.

Conclusion

Takeaway

Design rule

Use blue for structure and the proposed method, gold for one focal element, and charcoal for ordinary content.

Appendix

Appendix

Backup slide

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